

Packet Tracer - Propagate a Default Route in OSPFv2 (Instructor Version)

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only.

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.3.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	G0/0	172.16.2.1	255.255.255.0	N/A
	S0/0/0	172.16.3.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
	S0/1/0	209.165.200.225	255.255.255.224	
R3	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.10.6	255.255.255.252	N/A
	S0/0/1	192.168.10.10	255.255.255.252	
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Objectives

Part 1: Propagate a Default Route

Part 2: Verify Connectivity

Background

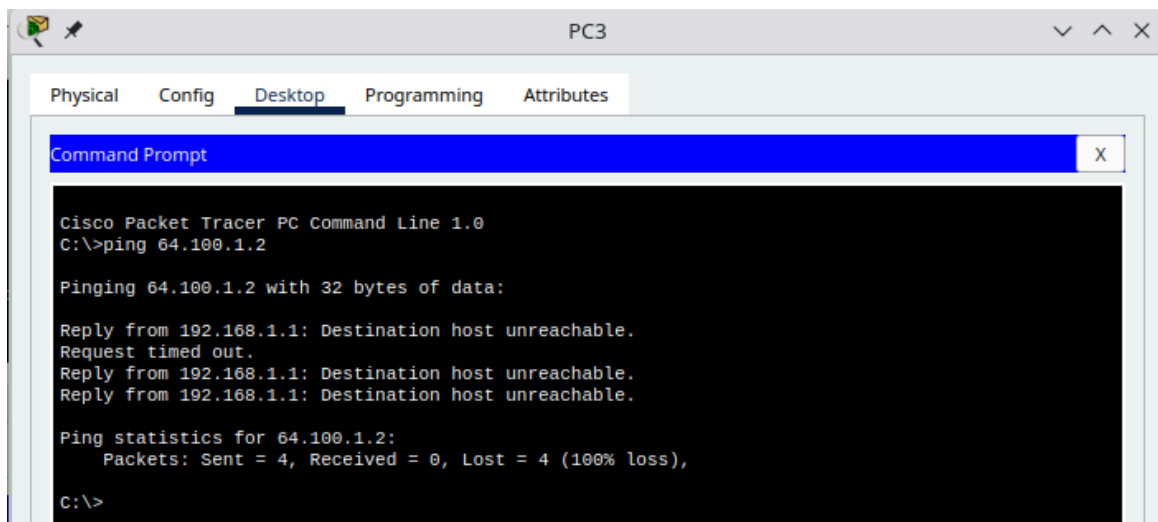
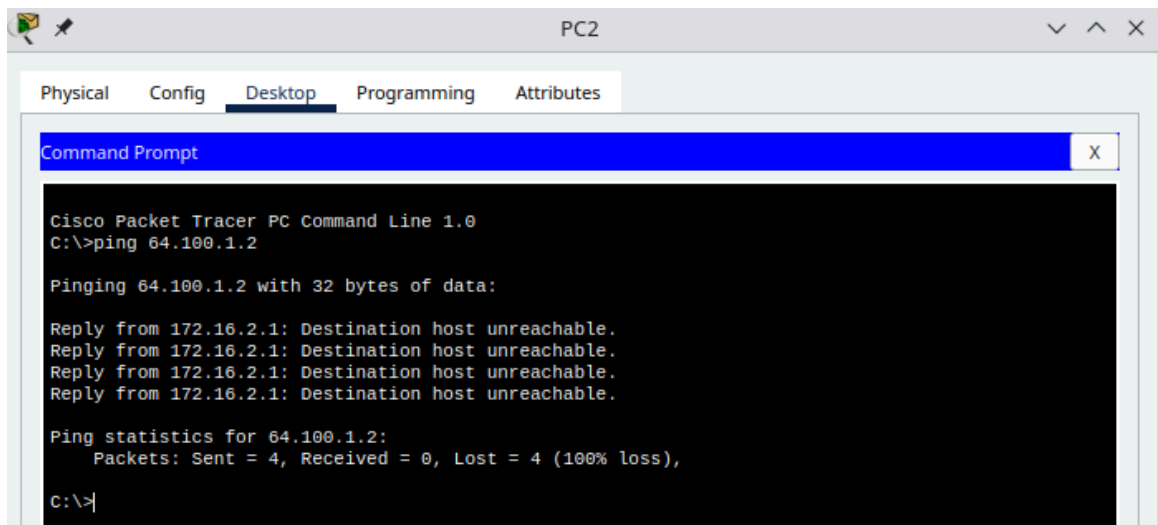
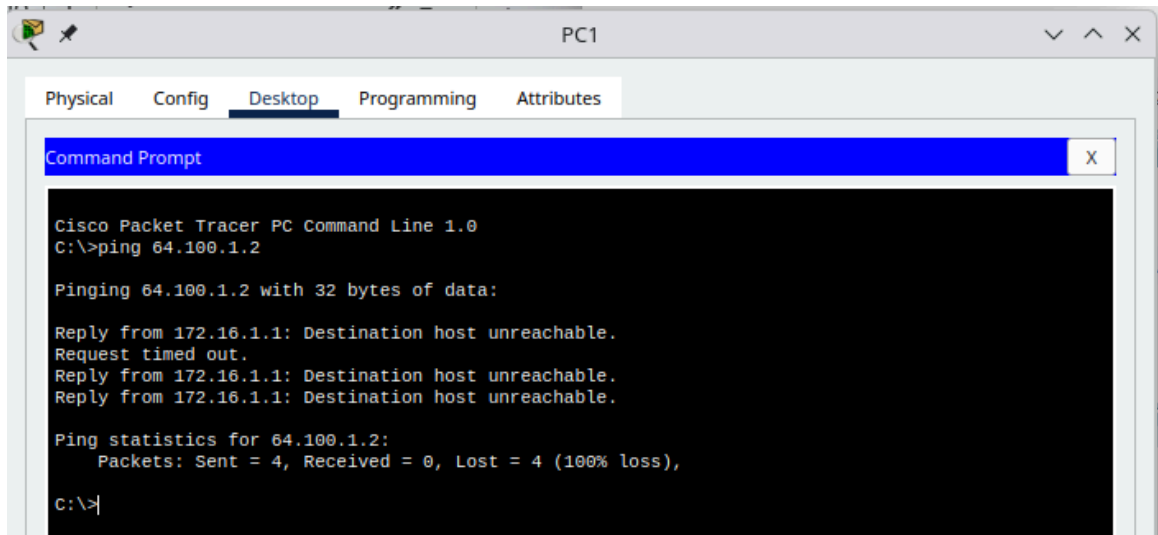
In this activity, you will configure an IPv4 default route to the Internet and propagate that default route to other OSPF routers. You will then verify the default route is in downstream routing tables and that hosts can now access a web server on the Internet.

Instructions

Part 1: Propagate a Default Route

Step 1: Test connectivity to the Web Server

- From PC1, PC2, and PC3, attempt to ping the Web Server IP address, 64.100.1.2.



Were any of the pings successful?

No.

What message did you receive, and which device issued the message?

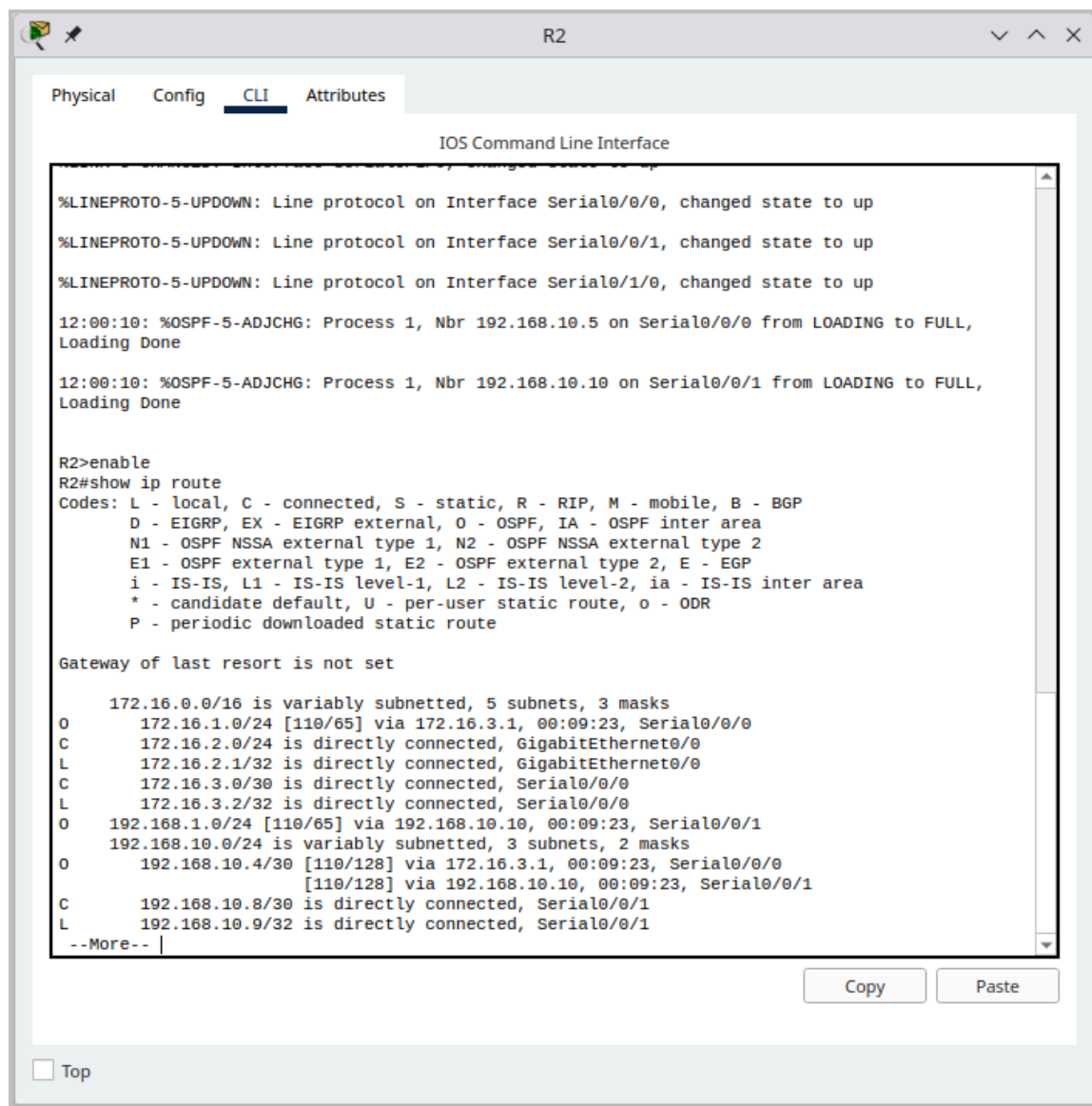
Destination unreachable from R2.

เพราะ R2 ซึ่งเป็นเราเตอร์ที่เชื่อมต่อกับอินเทอร์เน็ต (Web Server) ยังไม่มีเส้นทางที่จะไปยัง Web Server (หรือไปยังเครือข่าย 64.100.1.0/24) และ R2 เองก็ยังไม่ได้กำหนดค่า Default Route

- b. Examine the routing tables on routers R1, R2, and R3.

What statement is present in the routing tables that indicates that the pings to the Web Server will fail?

Gateway of last resort is not set



```
R2
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

12:00:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from LOADING to FULL, Loading Done
12:00:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.10 on Serial0/0/1 from LOADING to FULL, Loading Done

R2>enable
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
O   172.16.1.0/24 [110/65] via 172.16.3.1, 00:09:23, Serial0/0/0
C   172.16.2.0/24 is directly connected, GigabitEthernet0/0
L   172.16.2.1/32 is directly connected, GigabitEthernet0/0
C   172.16.3.0/30 is directly connected, Serial0/0/0
L   172.16.3.2/32 is directly connected, Serial0/0/0
O   192.168.1.0/24 [110/65] via 192.168.10.10, 00:09:23, Serial0/0/1
O   192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
O   192.168.10.4/30 [110/128] via 172.16.3.1, 00:09:23, Serial0/0/0
    [110/128] via 192.168.10.10, 00:09:23, Serial0/0/1
C   192.168.10.8/30 is directly connected, Serial0/0/1
L   192.168.10.9/32 is directly connected, Serial0/0/1
--More--
```

Step 2: Configure a default route on R2.

Configure R2 with a directly attached default route to the Internet.

```
R2(config)# ip route 0.0.0.0 0.0.0.0 Serial0/1/0
```

กำหนดเส้นทางหลัก (Default Route) บนเราเตอร์ R2 เพื่อให้รู้ว่าถ้าต้องการไปที่ไหนก็ตามที่ไม่รู้จัก (หรือเครือข่ายอื่น ๆ) ให้ส่งไปทางอินเทอร์เน็ตที่เชื่อมต่อกับอินเทอร์เน็ต

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 0.0.0.0 0.0.0.0 Serial0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R2(config)#ip route 0.0.0.0 0.0.0.0 Serial0/1/0
R2(config)#
```

ได้รับการแจ้งเตือนว่า "%Default route without gateway, if not a point-to-point interface, may impact performance"

Note: Router will give a warning that if this interface is not a point-to-point connection, it may impact performance. You can ignore this warning because it is a point-to-point connection.

Step 3: Propagate the route in OSPF.

Configure OSPF to propagate the default route in OSPF routing updates.

```
R2(config)# router ospf 1
R2(config-router)# default-information originate

R2(config)#router ospf 1
R2(config-router)#default-information originate
R2(config-router)#
```

Step 4: Examine the routing tables on R1 and R3.

Examine the routing tables of R1 and R3 to verify that the route has been propagated.

```
R1> show ip route
<output omitted>
Gateway of last resort is 172.16.3.2 to network 0.0.0.0
หมายความว่า R1 รู้แล้วว่าเกตเวย์สุดท้ายคือ R2 (ที่อยู่ IP 172.16.3.2) สำหรับการไปยังเครือข่ายใด ๆ (0.0.0.0)
<output omitted>
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:00:08, Serial0/0/0
O*E2 บอกว่านี่คือ External Route ประเภท 2 ที่เรียนรู้จาก OSPF
[110/1] คือ Administrative Distance (110) และ Metric (1) ของ OSPF
via 172.16.3.2 คือต้องส่งต่อผ่าน R2
Serial0/0/0 คืออินเทอร์เน็ตเฟซที่ใช้ไปหา R2
```

```
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
```

Gateway of last resort is 172.16.3.2 to network 0.0.0.0

```

    172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C    172.16.1.0/24 is directly connected, GigabitEthernet0/0
L    172.16.1.1/32 is directly connected, GigabitEthernet0/0
O    172.16.2.0/24 [110/65] via 172.16.3.2, 00:16:31, Serial0/0/0
C    172.16.3.0/30 is directly connected, Serial0/0/0
L    172.16.3.1/32 is directly connected, Serial0/0/0
O    192.168.1.0/24 [110/65] via 192.168.10.6, 00:16:31, Serial0/0/1
    192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
C    192.168.10.4/30 is directly connected, Serial0/0/1
L    192.168.10.5/32 is directly connected, Serial0/0/1
O    192.168.10.8/30 [110/128] via 192.168.10.6, 00:16:31, Serial0/0/1
    [110/128] via 172.16.3.2, 00:16:31, Serial0/0/0
    209.165.200.0/27 is subnetted, 1 subnets
O    209.165.200.224/27 [110/128] via 172.16.3.2, 00:16:31, Serial0/0/0
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:02:09, Serial0/0/0
```

R1>

!-----

R3> show ip route

<output omitted>

Gateway of last resort is 192.168.10.9 to network 0.0.0.0
เกตเวย์สุดท้ายของ R3 คือ R2 (ที่อยู่ IP 192.168.10.9)

<output omitted>

O*E2 0.0.0.0/0 [110/1] via 192.168.10.9, 00:08:15, Serial0/0/1

อธิบายคล้ายกับบน R1 แต่ช่องทางที่ใช้ไปหา R2 คือผ่าน Serial0/0/1 แทน

```

R3>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.9 to network 0.0.0.0

    172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O       172.16.1.0/24 [110/65] via 192.168.10.5, 00:17:47, Serial0/0/0
O       172.16.2.0/24 [110/65] via 192.168.10.9, 00:17:47, Serial0/0/1
O       172.16.3.0/30 [110/128] via 192.168.10.9, 00:17:47, Serial0/0/1
           [110/128] via 192.168.10.5, 00:17:47, Serial0/0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
    192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/0
L       192.168.10.6/32 is directly connected, Serial0/0/0
C       192.168.10.8/30 is directly connected, Serial0/0/1
L       192.168.10.10/32 is directly connected, Serial0/0/1
    209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 192.168.10.9, 00:17:47, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 192.168.10.9, 00:03:15, Serial0/0/1

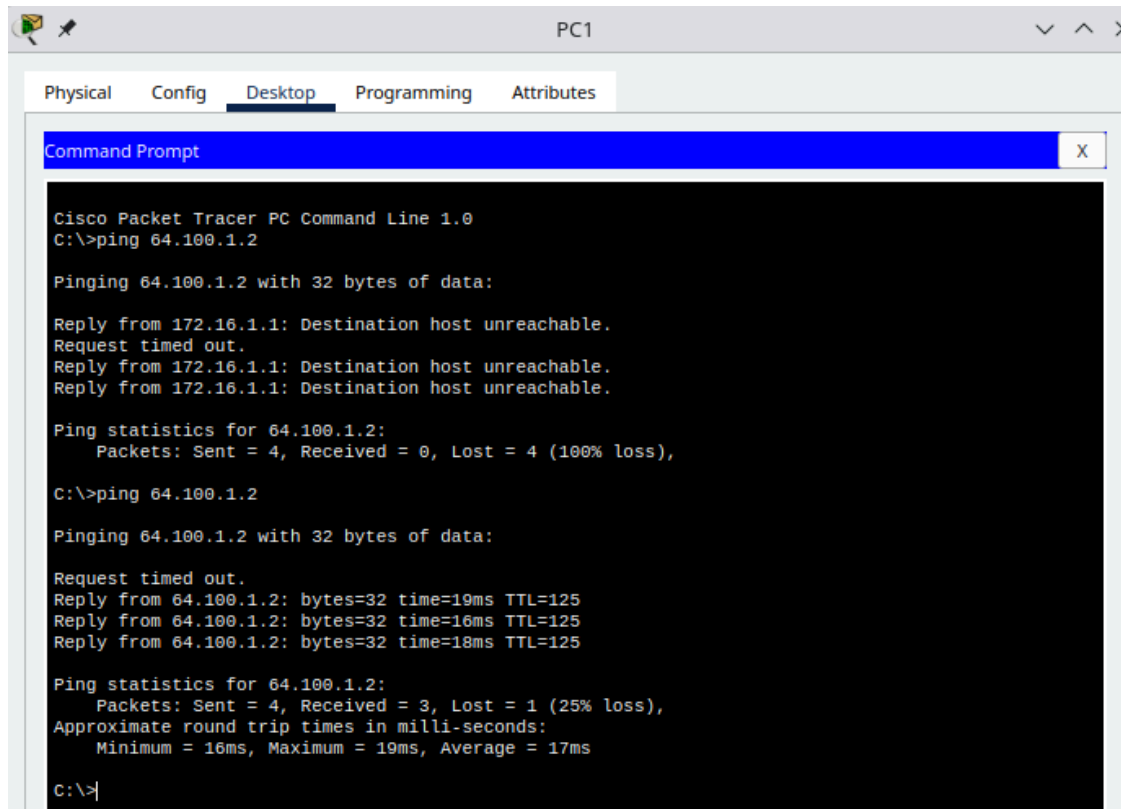
R3>

```

Part 2: Verify Connectivity

Verify that **PC1**, **PC2**, and **PC3** can ping the web server.

PC1 ping Web Server สำเร็จ



The screenshot shows a Cisco Packet Tracer interface for PC1. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The window title is 'Command Prompt' with a close button 'X'. The text inside the window shows the execution of a ping command to 64.100.1.2. The first attempt fails with 'Destination host unreachable' for all four packets. The second attempt succeeds for three out of four packets, showing round trip times of 19ms, 16ms, and 18ms, with a 25% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 172.16.1.1: Destination host unreachable.
Request timed out.
Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 64.100.1.2

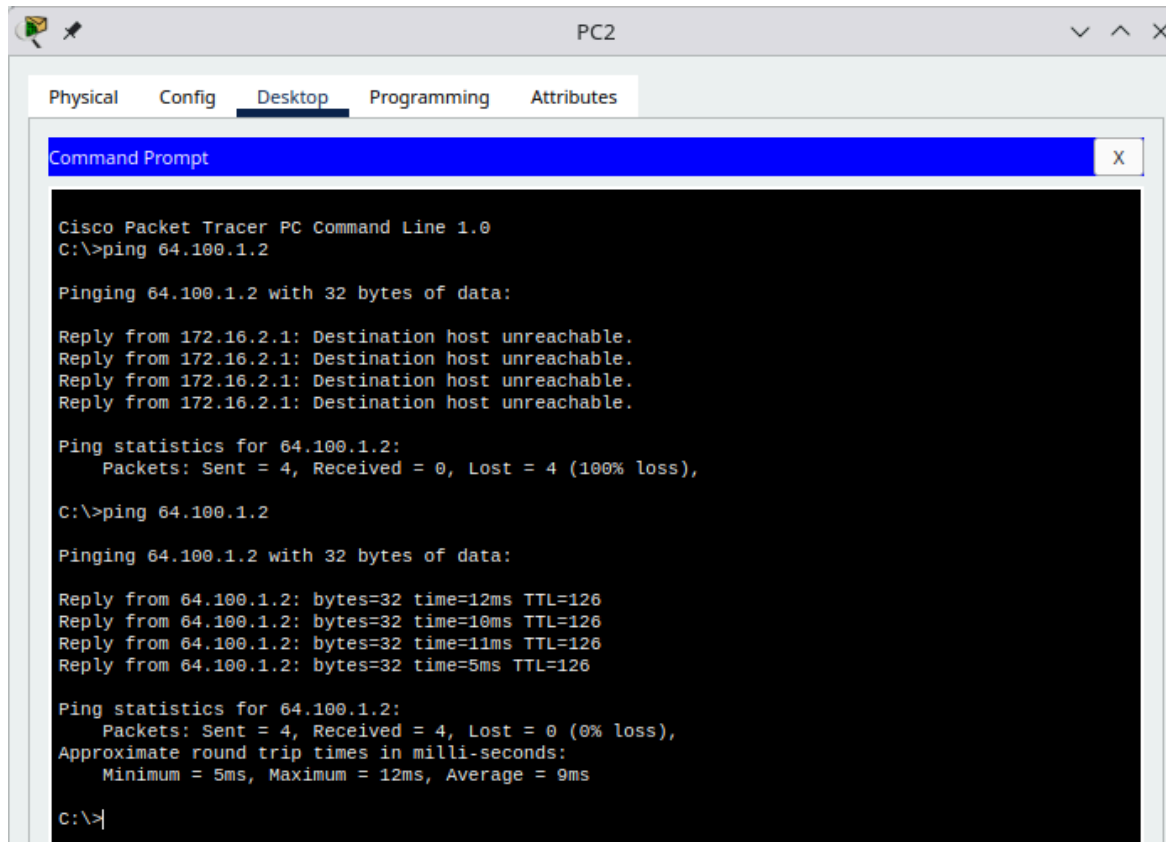
Pinging 64.100.1.2 with 32 bytes of data:

Request timed out.
Reply from 64.100.1.2: bytes=32 time=19ms TTL=125
Reply from 64.100.1.2: bytes=32 time=16ms TTL=125
Reply from 64.100.1.2: bytes=32 time=18ms TTL=125

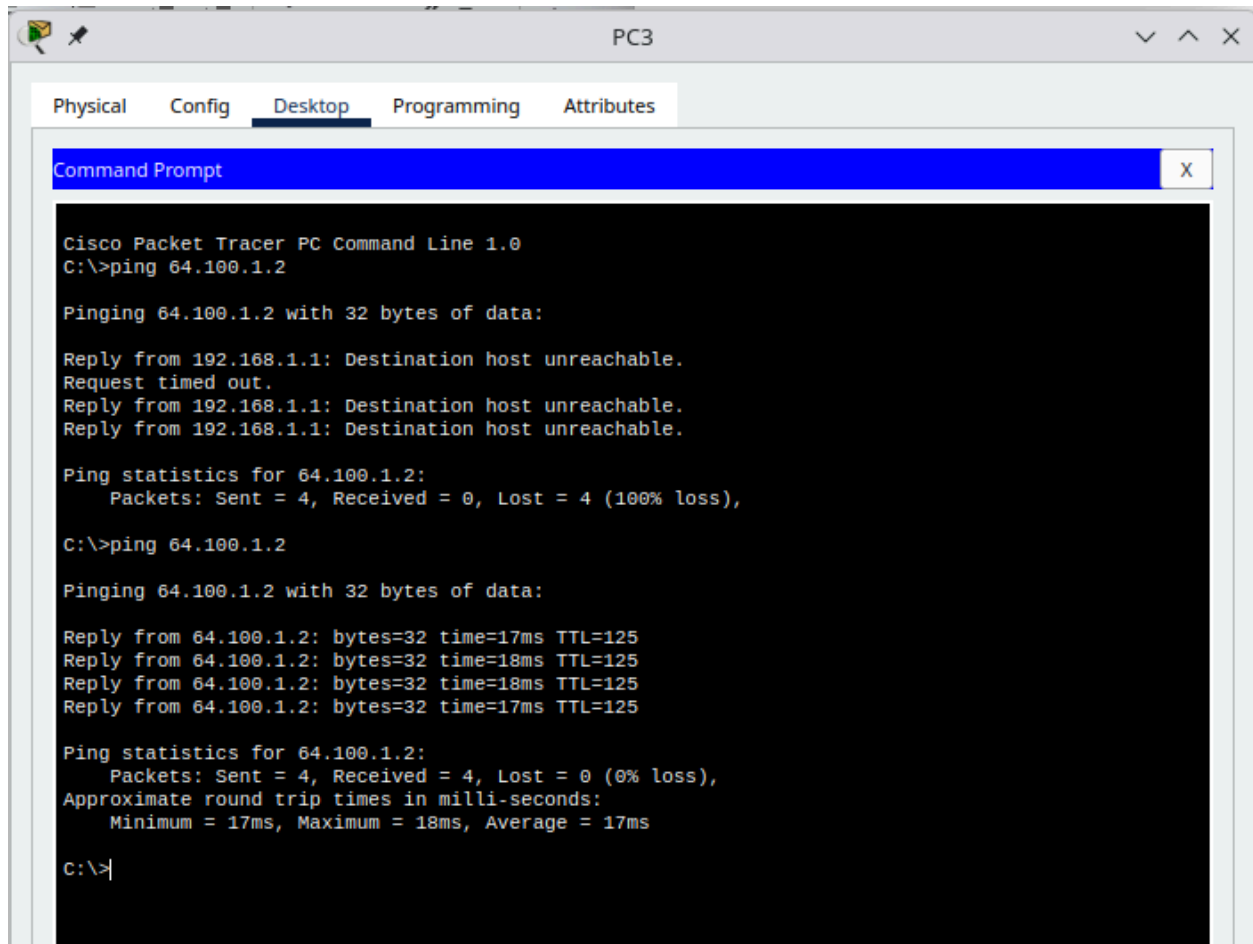
Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 19ms, Average = 17ms

C:\>
```

PC2 ping Web Server สำเร็จ



PC3 ping Web Server สำเร็จ



Answer script

Router R2

```
enable
config terminal
ip route 0.0.0.0 0.0.0.0 Serial0/1/0
router ospf 1
default-information originate
```